

CS501 Theory of Computation Comprehensive Revision Handbook

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Unit 1: Automata Theory, DFA & NFA

Automata Theory is the mathematical foundation of theoretical computer science.

A Deterministic Finite Automaton (DFA) is formally defined as a 5-tuple $M = (Q, \Sigma, \delta, q_0, F)$:

- Q : Finite non-empty set of internal states.
- Σ : Input alphabet containing finite distinct symbols (e.g. $\{0, 1\}$ or $\{a, b\}$).
- δ : Deterministic transition function mapping $Q \times \Sigma \rightarrow Q$. For every state and symbol, exactly one transition exists.
- q_0 : Initial/start state ($q_0 \in Q$).
- F : Subset of Q representing accepting or final states ($F \subseteq Q$).

Non-deterministic Finite Automata (NFA):

In an NFA, $\delta: Q \times \Sigma \rightarrow 2^Q$ (powersets of states). An NFA can transition to zero, one, or multiple states on a single symbol.

Equivalence: Every NFA has an equivalent DFA constructible via the Subset Construction (Powerset) algorithm. If an NFA has n states, its equivalent DFA has at most 2^n states.

Unit 2: Regular Expressions & Pumping Lemma

Regular Expressions (Regex) represent regular languages.

Key Identities:

- $R + S = S + R$
- $R + \Phi = R, R \cdot \epsilon = R$
- $R^* = (R + \epsilon)^*$
- $(R^*)^* = R^*$
- $(R + S)^* = (R^*S^*)^*$

Pumping Lemma for Regular Languages:

Let L be a regular language. There exists a pumping length $p \geq 1$ such that any string w in L with $|w| \geq p$ can be divided into three substrings $w = xyz$ satisfying:

1. $|y| \geq 1$ (the pumped part is non-empty)
2. $|xy| \leq p$ (the pump occurs within the first p characters)
3. For all $i \geq 0$, $x(y^i)z$ is in L .

Used systematically in university exams to prove that languages like $\{0^n 1^n \mid n \geq 0\}$ and $\{a^p \mid p \text{ is prime}\}$ are NOT regular.

Unit 3: Context-Free Grammars (CFG) & Pushdown Automata (PDA)

A CFG is defined as $G = (V, T, P, S)$:

- V: Finite set of variables (non-terminals).
- T: Finite set of terminal symbols.
- P: Production rules of the form $A \rightarrow \alpha$, where $A \in V$, $\alpha \in (V \cup T)^*$.
- S: Start symbol in V.

Chomsky Normal Form (CNF):

Every rule is either of the form $A \rightarrow BC$ or $A \rightarrow a$.

Pushdown Automata (PDA):

A PDA extends finite automata with a Last-In First-Out (LIFO) stack memory: $M = (Q, \Sigma, \Gamma, \delta, q_0, Z_0, F)$.

Stack operations: push, pop, and no-op (replace top symbol). Deterministic PDAs (DPDA) accept a strict subset of Context-Free Languages (Deterministic CFLs).

Unit 4: Turing Machines & Decidability

Turing Machine (TM) is the universal computational model: $M = (Q, \Sigma, \Gamma, \delta, q_0, B, F)$.

The transition function $\delta(q, X) = (p, Y, D)$ updates the current state to p, writes symbol Y on the infinite tape, and shifts the read/write head in direction D in {L, R}.

Halting Problem (Turing, 1936):

Given the description of an arbitrary Turing machine M and input w, determining whether M halts on w is undecidable. Proved using Cantor's diagonalization technique. Post Correspondence Problem (PCP) is also undecidable and commonly tested in university exams.

